



Name \_\_\_\_\_

| ID number | Date | Group |
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Signature

Select one option per question. You can justify the answer that you select. **(6 points each question.)**

- Which of the following options evaluates correctly  $(f \circ g)(x)$ 
  - $g(f(x))$
  - $g(x)f(x)$
  - $f(g(x))$
  - $f(x)g(x)$
- Which of the following functions models exponential decay.
  - $m(t) = Ae^{-r+t}$
  - $m(t) = -Ae^{r-t}$
  - $m(t) = Ae^{-rt}$
  - $m(t) = -Ae^{-rt}$
- Considering  $f(x) = (x + 1)^2 - 1$ . What would the new function be like after shifting 5 units to the left and 3 units down?
  - $f(x) = (x + 5)^2 - 3$
  - $f(x) = (x + 2)^2 - 4$
  - $f(x) = (x + 2)^2 + 2$
  - $f(x) = (x + 6)^2 - 4$
- Given the function  $f(x) = -2\sqrt{x^2 - 4} - 5$ , which of the following represents its domain?
  - $-\infty < x < \infty$
  - $-\infty < x \leq 4$
  - $-\infty < x \leq 0$
  - $2 \leq x < \infty$

1) Solve the following equations. Simplify the answer as much as possible and check that all solutions satisfies the original equation. **(6 points each equation)**

a.  $\log(3-x) + \log(x+9) = 0$

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b.  $6^{x-3} - 2^{x+2} = 0$

## Algebraic and Transcendental Functions

- 2) For the functions  $f(x) = \frac{5x}{x+2}$  and  $g(x) = \frac{8}{x}$  find the indicated composition and reduce the expression as much as possible, **(6 points each operation)**

a.  $(f \circ g)(x)$

b.  $(g + g)(x)$

- 3) Use logarithmic properties to condense the following expression  $2 \log_8 y + \frac{1}{3}(\log_8 x - 2 \log_8 w)$ .  
(6 points.)

Section 3: Function analysis (12 points)

- 1) Find the inverse function of  $f(x) = \frac{2x-4}{3x-10}$ . Reduce the expression as much as possible and proof your answer by evaluating the function composition  $(f \circ f^{-1})(x) = x$ . **(6 points)**

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- 2) Find the roots of the function  $f(x) = \frac{x^2+x-2}{x+2}$ . **(6 points)**

Section 4: Modeling with math (30 points)

- 1) Hydrogen potential (pH) is a number used to describe the acidity or basicity of a chemical substance and is defined by the equation  $pH = -\log(H^+)$ , where  $H^+$  measures the concentration of hydrogen ions moles per liter.
- i. Determine the  $pH$  of a coke if  $H^+ = 1 \times 10^{-3}$ . **(6 points)**
  - ii. What is the hydrogen ions concentration of the stomach acid with  $pH$  of 2?. **(6 points)**

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- 2) The half-life of Radium-266 is 1600 years. Suppose that there is a sample of 22mg. Recalling that the mass decays exponentially, fulfill the following points
- i) Write the function that models the remaining mass after  $t$  years. **(6 points)**
  - ii) How much mass will remain after 4000 years? **(6 points)**
  - iii) How much time will it take for the sample to be only a quarter of its initial value?. **(6 points)**

Section 5: Conic Sections (4 points)

- 1) Identify the conic section given the following general equation  $16x^2 - 9y^2 - 96x - 54y - 81 = 0$ .  
**(4 points.)**